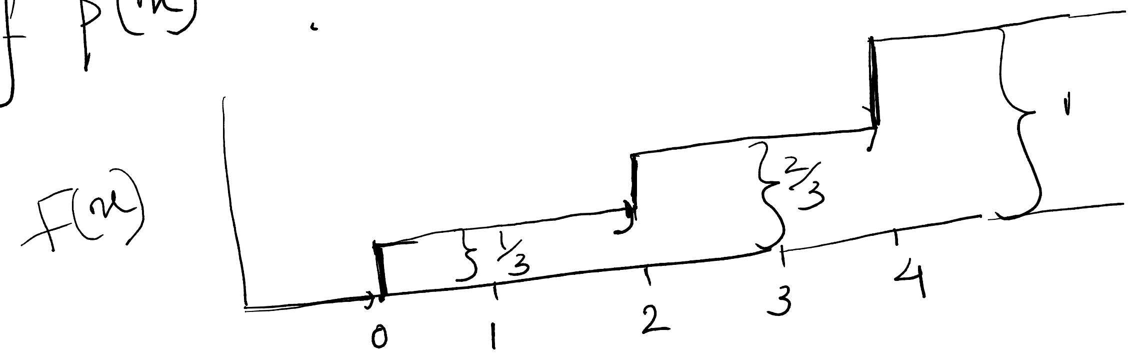
PRST Class (14/10/2020)

tutorial - 2

Q) $F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{3} & 0 \le x < 2 \\ \frac{2}{3} & 2 \le x < 4 \\ 1 & x \geq 4 \end{cases}$

pmf $p(x) = ?$



$F(x) = P(X \le x)$

$P(X \le 0) = \frac{1}{3} = F(0)$

$P(X < 0) = 0$

$P(X \le 0) - P(X < 0) = P(X = 0) = \frac{1}{3}$

~~F(x~~ $F(2) = \frac{2}{3}$

$P(X \le 2) = \frac{2}{3}$

$P(X < 2) = \frac{1}{3}$

$P(X \le 2) - P(X < 2) = P(X = 2) = \frac{1}{3}$

$F(4) = 1$

$P(X \le 4) = 1, \quad P(X < 4) = \frac{2}{3}$

$P(X \le 4) - P(X < 4) = 1 - \frac{2}{3}$

$= \frac{1}{3}$

X	$P(X = x)$
0	$\frac{1}{3}$
2	$\frac{1}{3}$
4	$\frac{1}{3}$

4	$\frac{1}{3}$

7).

X	$P(X=x)$
0	0.3
2	0.1
3	0.6
	1

$$Y = 3(X-1)^2.$$

$$E(g(X)) = \sum_x g(x) P(X=x)$$

$$E(Y) = \sum_{x=0,2,3} 3(x-1)^2 P(X=x)$$

$$= (3 \times 0.3 + 3 \times 0.1 + 12 \times 0.6)$$

6).

$$f(x,y) = \left(x^2 + \frac{xy}{3}\right). \quad 0 < x < 1, \; 0 < y < 2$$

$$P\left(Y \leq \frac{1}{2}\right).$$

$$f_Y(y) = ?$$

$$f_Y(y) = \int_0^1 f(x,y)\, dx$$

$$= \int_0^1 \left(x^2 + \frac{xy}{3}\right) dx$$

$$= \left[\frac{x^3}{3} + \frac{y}{3}\frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{3} + \frac{1}{6} y$$

$$f_Y(y) = \frac{1}{3} + \frac{1}{6} y \; ; \quad 0 < y < 2$$

$$P\left(Y \leq \frac{1}{2}\right) = \int_0^{1/2} f_Y(y)\, dy$$

$$= \int_0^{\frac{1}{2}} \left(\frac{1}{3} + \frac{1}{6} y\right) dy$$

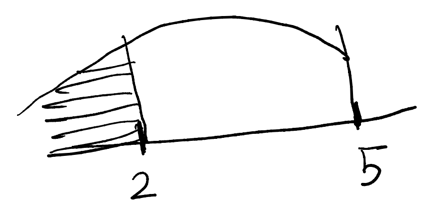
$$\left[\quad \frac{y^2}{} \right]_0^{\frac{1}{2}}$$

$$= \left[\frac{1}{3} y + \frac{y^2}{12}\right]_0^2$$

$$= \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{4 \cdot 12}$$

5) $F(x) = 1 - e^{-2x}, \quad x > 0$

$P(2 \le X \le 5) = ?$



$P(2 \le X \le 5)$

$$= P(X \le 5) - P(X \le 2)$$

$$= F(5) - F(2)$$

$$= \left(1 - e^{-2\times 5}\right) - \left(1 - e^{-2\times 2}\right)$$

$$= e^{-4} - e^{-10}$$

$$P(X \le 2) - P(X < 2) = P(X = 2) = 0$$

$$P(X \le 2) = P(X < 2)$$

$$F(2) \qquad F(2-\varepsilon)$$

$$f(x) = \frac{dF(x)}{dx}$$

$$= 2e^{-2x}, \quad x > 0$$

4)
$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

$$f(x) = \begin{cases} 0 & x < a \\ \frac{1}{b-a} & a \le x \le b \\ 0 & x > b \end{cases}$$

$$E(X) = \int_a^b x \cdot \frac{1}{b-a} \, dx$$

$$(b^2 - a^2) \qquad (b+a)$$

$E(X)$ ... a

$= \frac{1}{b-a} \frac{(b^2-a^2)}{2} = \frac{(b+a)}{2}$

$V(X) = E(X^2) - [E(X)]^2$

$E(X^2) = \int_a^b x^2 \frac{1}{b-a} \, dx$

$= \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3}$

$$f(x) = \begin{cases} 0 & x < a \\ \frac{1}{b-a} & a \le x \le b \\ 0 & x > b \end{cases}$$

$F(x) = \int_{-\infty}^{a} f(u)\, du$ $\quad x < a$

$= 0$

$F(x) = \int_{-\infty}^{x} f(u)\, du$ $\quad a \le x \le b$

$= \cancel{\int_{-\infty}^{x} \frac{1}{b-a} du}$

$= \int_{-\infty}^{a} f(u)\, du + \int_a^x f(u)\, du$

$= 0 + \int_a^x \frac{1}{b-a} du$

$= \frac{x-a}{b-a}$

$F(x) = \int^{x} f(u)\, du$ $\quad x > b$

$$F(x) = \int_{-\infty}^{x} f(u)\,du \qquad x > b$$

$$= \int_{-\infty}^{a} f(u)\,du + \int_{a}^{b} f(u)\,du + \int_{b}^{x} f(u)\,du$$

$$= 0 + \int_{a}^{b} f(u)\,du + 0$$

$$= 0 + \frac{b-a}{b-a} + 0$$

$$= 1$$

10.

	$Y=1$	$Y=2$	$Y=3$	
$X=1$	$1/6$	0	$\frac{1}{6}$	$\frac{1}{3}$
$X=2$	$a_{21} = 0$ $\frac{1}{3} - \frac{1}{12} - \frac{1}{4}$	$1/4$	$a_{23} = \frac{1}{12}$ $= \frac{1}{2} - \frac{1}{6} - \frac{1}{4}$	$\frac{1}{3}$
$X=3$	$a_{31} = 0$ $= \frac{1}{6} - 0 - \frac{1}{6}$	a_{32} $= \frac{1}{3} - \frac{1}{4}$	$1/4$	$\frac{1}{3}$
	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	1